Nikolas Torosian

Cover Letter

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Hi!

I do extreme sports, trips in the wild and read the BSD's man pages. Currently working on an electronic musical instrument company, targeting a specific format called Eurorack Synthesizers. I will be happy to talk for any of these, but below are important experience and experties.

Employment

- **Founder of electronic instruments production laboratory**
  10/2025 until now, Awfulbytes
  The main goal of our space is to construct and supply open hardware/software electronic instruments. For more information you could visit www.awfulbytes.org or contact us with any questions at info@awfulbytes.org.

- **Practical work semester**
  11/2023 until 04/2024, S.C.O CommonsLab
  Most of the time spend for my mandatory practical work for the Bachelor of Mechanical Engineering was about designing and constructing with 3D printing plastic, cases for electronic devices ment for industrial environment.

- **N.C. Press Brake Technician**
  06/2022 until 12/2022, Mataliotakis A.E. Heraklion, Crete
  Most of my time there was custom solutions for customers with a variety of needs and specifications. Most notably we were supplying a local truck mechanic shop with folded sheet metal compartments. Among other, parts for some race cars where made there with the owners approval.

Conferences

Oral presentations

- **Filtering approaches for wind tunnel power measurements in an inverter induced noise environment**
  1 June 2022
  DEMSEE-2022 14th Deregulated Electricity Market issues in South-Eastern Europe
  Nikolaos Papadakis, Konstantine Kontaksakis, Nikolaos Torosian

Research Experience

- **Research of Direct Digital Synthesis for a tunable oscillator in C**
  08/2024 until now
  For the construction of a tunable voltage controlled oscillator programmed on an ARM mcu with fixed point arithmetic, a huge number of designes (theoretical consepts and applyied algorithms) where studied and prototyped for the construction of the first product for Awfulbytes.

- **Design development and production of an electronic devices' housing**
  11/23 until 04/24
  Practical work semester at CommonsLab
  From design to production of the housing was all inhouse with 3D printing manufacturing. Many difficulties were delt with early in development of engineering samples. The project waw funded by the E.U., in a program called Better Factory, aiming to upscale and modernize the automation technologies in factories.

- **Digital Signal Post-Processing and noise reduction from EMI**
  2021 until 2023
  Hellenic Mediterranean University
  Analisys of linearly recorded signals and filtering of electromagnetic interferences from the environment. All the work is available on github here

Professional Skills and experties

- **CAD/CAM/CAE/ECAD Software**
  Knowledgable in all software from Autodesk, with speciality at Inventor, Fusion360, AutoCad. General knowledge about PTC Creo CAM capabilities and realy good knowledge on circuit and pcb design primarily on KiCad. Extended knowledge about printed circuit board behavior and analysis.

- **C Programming Language**
  Very good knowledge on microcontroller bare metal programming with experties in the ARM family, and extensive familiarity with freestanding applications. Less knowledgable about real-time OS's like Zephyr/RTOS.

- **LaTeX**
  Advanced knowledge of compiling technical reviews and documents. What you are reading now is in LaTeX and you can confirm it here.

- **Emacs text editor**
  Excellent knowledge of the base installation with personallized settings to enhance productivity. My dissertation for the Bachelor as well as most of the documents I produced were heavily dependent in Org Mode and written in the Org Rich Text format. The ability to run code blocks and produce tables at the time of compiling to pdf, gave me the oportunity to focus on improving the material and phrasing of the document instead of hunting for my latest .png plot.

- **Zig Programming Language**
  Good level of understanding and using the features of the language, like arena allocators and scope defer. Mostly worked on TUI's with simple but interesting command line interface tools.

- **Rust Programming Language**
  Good knowledge of the standard library and external libraries for signal analysis and plot generation from datasets. Adequate knowledge of the overall ecosystem of the language.

- **Python Programming Language**
  Intermediate knowledge of the language and increased knowledge on the overall ecosystem of the language. Signal analysis and interactive puzzle game using QR codes as pieces are among projects I have worked on.

- **Github Actions**
  Good knowledge of the tools offered by github for CI/CD. Used to test python projects in Windows and Linux, briefly used to generate simple PDF technical documents.

Education

- **Bachelor in Mechanical Engineering T.E.**
  2024, Hellenic Mediterranean University

- **High School Education**
  2013–2014, Erasmios Greek & German School Athens, Greece